

# Networks Mini-Project: The All-Or-None model

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Preferential attachment models help in the understanding of network growth. In this report we will investigate the all-or-none model variation by analytically deriving key metrics, verifying them through numerical simulations. The created network undergoes a phase change at  $p = 0.5$ , changing from a sparse to a dense regime. Both estimates for the clustering coefficients align, exhibiting a non-linear scaling as the network transitions from a random tree graph at  $p = 0$  to a fully connected graph when  $p = 1$ . Finally, the average shortest-path distance decreases linearly as  $p$  increases due to the introduction of topological shortcuts.

## I. INTRODUCTION

Preferential attachment and copying network models are paradigmatic mechanisms for network growth. The study of such systems helps us to understand how complex networks evolve in the real world. In this report, we will consider a variation of such models, the all-or-none model. At time  $t = 0$  we begin with a single node. At each subsequent time step, a new node is added and connected to an existing node chosen uniformly at random. Further, the new node attempts to connect to all the neighbours of the target node. With probability  $p$ , it connects to all neighbours, and hence, with probability  $1 - p$ , it connects to none. In this report, we investigate the structural properties of the resulting network, analytically deriving key structural metrics which we will verify through numerical simulations. This, in turn, highlights the effect  $p$  has on the resulting networks.

## II. EDGE AND DEGREE DISTRIBUTION ANALYSIS

We begin our discussion by investigating how the number of edges in a network produced by the all-or-none model varies. First, consider the expected number of edges  $E$  of a network of size  $N$  with a connection probability  $p$ , given by

$$E_N - E_{N-1} = 1 + p\langle k \rangle_{N-1} = 1 + p\left(\frac{2E_{N-1}}{N-1}\right) \quad (1)$$

where  $\langle k \rangle_N$  is the average degree of nodes at network size  $N$ . The first term on the right-hand side of Eq. (1) arises since at every time step we add one edge between the new node and the target node regardless of parameters, whereas the second term arises since, with probability  $p$ , the new node connects to all the target nodes' neighbours. Taking the limit of  $N \rightarrow \infty$ , using the fact  $t = N - 1$ , we get the continuum approximation

$$E(t) = \begin{cases} \frac{1}{1-2p}(t - 2pt^{2p}) & \text{if } p \neq 0.5 \\ t(\ln(t) + 1) & \text{if } p = 0.5 \end{cases} \quad (2)$$

Whilst Eq. (2) provides a good approximation for large  $N$  we can also find a closed form solution to the recurrence relation. By introducing a factor  $f(N) = \prod_{k=2}^N \frac{k-1+2p}{k-1} = \frac{\Gamma(N+2p)}{\Gamma(N)\Gamma(1+2p)}$  and applying the condition  $E_1 = 0$ , we can

exploit the properties of the gamma function to find that

$$E_N = \begin{cases} \frac{1}{1-2p}(N - f(N)) & \text{if } p \neq 0.5 \\ N \sum_{k=2}^N \frac{1}{k} & \text{if } p = 0.5 \end{cases} \quad (3)$$

From Fig. 1, we can see that both theoretical mod-

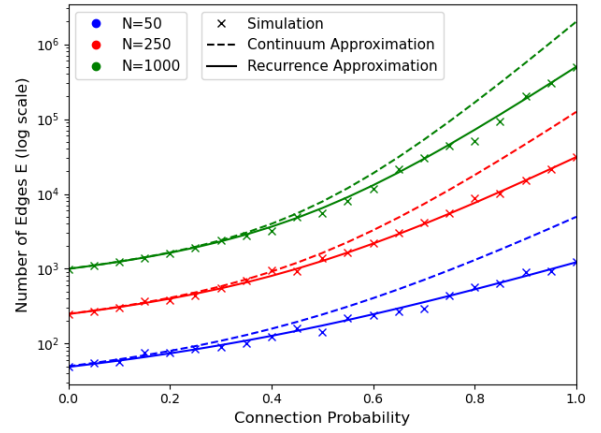


FIG. 1. The number of edges  $E$  as a function of the connection probability  $p$  for various system sizes  $N$ .

els capture the growth of  $E$ , as a function of  $p$  and  $N$ . The continuum approximation tends to overestimate  $E$ , which is expected due to the finite size of  $N$  and the discrete nature of the simulation. The dominant term in the continuum approximation, Eq. (2), is linear in  $N$  in the sparse regime,  $p \in [0, 0.5)$ , and in the dense regime,  $p \in (0.5, 1]$ , proportional to  $N^{2p}$ . Furthermore, since the average degree scales as  $\frac{2E}{N}$ , this non linearity in the dense regime implies that the average degree diverges, output structurally denser networks as  $N$  increases. Suggesting a phase change at the critical value  $p = 0.5$ , dictating a more rapid growth in  $E$ . At higher values of  $N$  the simulated data also shows this trend, whilst to lesser extent. This matches the description of the model as once  $p$  crosses such threshold, a newly added node is more likely than not to connect to the neighbours of the target node, driving the non-linear increase in  $E$ . Whilst it is far less trivial to derive such information from Eq. (3), in Fig. 1 we observe it models the same properties to less of an extreme. It models such transition with greater accuracy, as can be seen with how closely it matches the simulated data.

Turning our attention to degree distribution of the model in the sparse regime,  $p \in [0, 0.5)$ . The attachment

probability is the probability that a new node forms a link with a specific node of degree  $k$ , given by

$$\Pi(k) = \frac{1}{N-1} + p \frac{k}{N-1} \quad (4)$$

The first term of Eq. (4) is the uniform probability that the node will be selected as the target node, and the second term is the probability that the node is a neighbour of the target node, in which case a link occurs with probability  $p$ . We assume  $N$  is large and employ a continuum (rate equation) approach. Let  $N_k$  be the expected total number of nodes of degree  $k$ , with the corresponding density  $n_k = \frac{N_k}{N}$ . The rate equation for the time evolution of  $N_k$  is

$$\frac{d}{dN} N_k = \Pi(k-1)N_{k-1} - \Pi(k)N_k + \delta_{k,1}(1-p) + \gamma_k \quad (5)$$

where the first term accounts for nodes of degree  $k-1$  becoming nodes of degree  $k$ , the second accounts for nodes of degree  $k$  ceasing to be nodes of degree  $k$ , and the third term accounts for new nodes entering the system with degree 1. The fourth term accounts for the new node entering the system with degree  $k$ , where  $k = 1 + k_t$  with  $k_t$  being the degree of the target node. For  $N$  large  $\gamma_k$  becomes negligible, and we can take  $N-1 \rightarrow N$  in the formula for  $\Pi$ , implying that

$$N \frac{d}{dN} n_k = (1+p(k-1))n_{k-1} - (2+pk)n_k + \delta_{k,1}(1-p) \quad (6)$$

In the sparse regime we have shown the number of edges grows linearly with  $N$ , implying that, since  $\langle k \rangle_N = \frac{2E}{N}$ , the average degree converges to a finite constant. Since the average degree is stable, it is reasonable to assume that the densities  $n_k$  asymptotically converge to a stationary solution, giving the relation

$$(2+pk)n_k = (1+p(k-1))n_{k-1} + \delta_{k,1}(1-p) \quad (7)$$

which, when solved, gives the distribution

$$p(k) \propto k^{-\frac{1+p}{p}} \quad (8)$$

To analyse the behaviour in Fig. 2, we examine the ratio  $\frac{n_k}{n_{k-1}} = \frac{1-p+pk}{2+pk}$  derived from Eq. (7). For small  $k$ , the uniform attachment probability dominates  $\Pi$ , leading to  $n_k \approx 0.45n_{k-1}$ . This corresponds to the initial exponential decay in  $p(k)$ . On a log-log scale graph, this is displayed by the initial concave downward curvature where the recurrence relation models the simulated data to high accuracy. As  $k$  increases, preferential attachment will become more dominant, leading to a small fraction of nodes gaining the vast majority of links, which in turn causes the initial exponential decay to flatten into a heavy tail. Whilst the power law sits above the simulated data since it represents the distribution in the limit of  $k \rightarrow \infty$ , Fig. 2 shows that the simulated data closely follows this linear trajectory on the logarithmic scale. Further, the recurrence relation also matches this trend whilst tending to

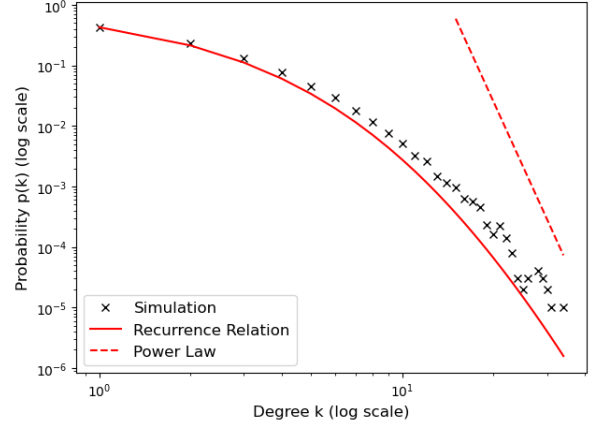


FIG. 2. The degree distribution for network size  $N = 10000$  and connection probability  $p = 0.1$ .

underestimate, due to the nature of its derivation in which we assumed  $N \rightarrow \infty$ . We expected that as  $N$  increases, the recurrence relation will better match the simulated graph, and that in the limit of  $k \rightarrow \infty$  the power law will perfectly match the theoretical trajectory. Finally, for our specific example of  $p = 0.1$ , the pronounced curvature arises because the preferential attachment term is weakly weighted, which in turn allows uniform attachment to dominate a large portion of the network's growth. As  $p$  increases, the model places more weight on preferential attachment, shortening the exponential regime and producing a much flatter, thicker tail. This structural shift is mathematically reinforced by the decrease in the scaling exponent  $\gamma = \frac{1+p}{p}$ .

### III. CLUSTERING AND ASSORTATIVITY

To explore clustering we begin by considering the number of triangles produced by our model. Let  $T_N$  be the expected number of triangles at network size  $N$ , then

$$T_N = T_{N-1} + p(\langle k \rangle_{N-1} + \langle T_i \rangle_{N-1}) \quad (9)$$

$$= T_{N-1} + p\left(\frac{2E_{N-1}}{N-1} + \frac{3T_{N-1}}{N-1}\right) \quad (10)$$

where  $T_i$  is the number of triangles containing node  $i$ . The first term arises since our new node forms a triangle with the target node and each of the target node's neighbours, and the second term arises due to the fact the new node forms a triangle with each pair of neighbours of the target node that have a direct link, both occurring with probability  $p$ . Hence we take the average of both to form an equation for the expected number of triangles. This directly gives us a recurrence relation for  $T_N$ . We also consider the continuum approximation, which in the limit  $N \rightarrow \infty$ , after solving a differential equation and applying the initial condition  $T(2) = 0$  ( $N = 2, t = 1$ ),

giving that

$$T(t) = \begin{cases} \frac{2p}{1-2p} \left( \frac{1}{1-3p} t + 2t^{2p} \right) - \frac{6p}{1-3p} t^{3p} & \text{if } p \neq 0.5, \frac{1}{3} \\ 2t(3t^{0.5} - 3 - \ln(t)) & \text{if } p = 0.5 \\ 2(t(\ln(t) - 2) + 2t^{2/3}) & \text{if } p = \frac{1}{3} \end{cases} \quad (11)$$

Fig. 3 shows the high level of accuracy provided by the

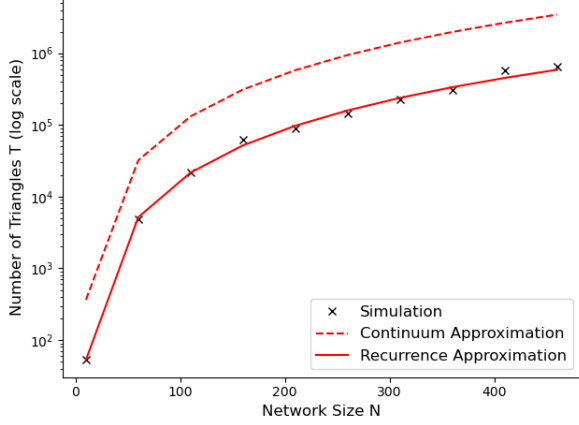


FIG. 3. The number of triangles  $T$  as a function of network size  $N = 1000$  for  $p = 0.75$ .

discrete recurrence relation for  $T$ . Although the deviation of the continuous approximation from the simulated data grows as  $N$  increases, it successfully captures the scaling behaviour of  $T$ , highlighted when plotting on a logarithmic scale.

Instinctively, the next thing to consider is the clustering coefficient

$$C = \frac{3 \times \text{Number of Triangles}}{\text{Number of Connected Triplets}} \quad (12)$$

Let  $R_N$  be the expected number of triplets for network size  $N$ , we have

$$R_N = R_{N-1} + \langle k \rangle_{N-1} + \frac{p}{2} (\langle k^2 \rangle_{N-1} + \langle k \rangle_{N-1}) + p \langle k^2 \rangle_{N-1} \quad (13)$$

where the first term arises since the new node always connects with the target node, thus forming a triplet with each of the target node's neighbours and the target node itself. The second term represents how many triplets are formed with the new node at the centre. Let  $u$  denote the new node and  $v$  denote the target node. The new node is centred at  $\binom{k_u}{2}$  triplets and thus, with probability  $p$ , since  $k_u = 1 + k_v$  and  $\binom{k_u}{2} = \frac{k_u(k_u-1)}{2}$  we reach the conclusion for the second term by taking expectations, noting that no triplets have  $u$  at their centre if connection doesn't occur. The third term represents how many triplets are formed with the new node and a neighbour of the target node, say  $w$ , at the centre. Again with probability  $p$ , the number of triplets formed in this manner will be  $\sum_{w \in N(v)} k_w$  where  $N(v)$  denotes the neighbourhood of  $v$ . Averaging over all

target nodes gives

$$\frac{1}{N-1} \sum_{v=1}^{N-1} \sum_{w \in N(v)} k_w = \frac{1}{N-1} \sum_{w=1}^{N-1} k_w^2 = \langle k^2 \rangle_{N-1} \quad (14)$$

where we have used the fact that  $w \in N(v)$  exactly  $k_w$  times, hence our formulation for  $R_N$ . Using the fact that

$$\langle k^2 \rangle_{N-1} = \frac{2}{N-1} (R_{N-1} + E_{N-1}) \quad (15)$$

we get

$$R_N = R_{N-1} \left( 1 + \frac{3p}{N-1} \right) + E_{N-1} \left( \frac{2+4p}{N-1} \right) \quad (16)$$

In the limit  $N \rightarrow \infty$ , we get the following continuum approximation

$$R(t) = \begin{cases} \frac{2+4p}{1-2p} \left( \frac{1}{1-3p} t + 2t^{2p} \right) - \frac{6+12p}{1-3p} t^{3p} & \text{if } p \neq 0.5, \frac{1}{3} \\ 8t(3t^{0.5} - 3 - \ln(t)) & \text{if } p = 0.5 \\ 10(t(\ln(t) - 2) + 2t^{2/3}) & \text{if } p = \frac{1}{3} \end{cases} \quad (17)$$

from which we can derive our first estimate of the clustering coefficient, made easy by the fact that  $T(t) = cR(t)$ , as seen by comparing Eq. (17) and Eq. (11), for all  $p$ , where  $c$  is a constant or function of  $p$ , giving that in the limit  $N \rightarrow \infty$

$$C = \frac{3p}{1+2p} \quad \forall p \in [0, 1] \quad (18)$$

which we notice is now continuous for all  $p \in [0, 1]$ . In addition, it is clear that the only difference in the recurrence relations Eq. (16) and Eq. (10) for  $R_N$  and  $T_N$  is the prefactor of  $E_{N-1}$ . Furthermore, since both have the same initial condition, that is  $T_1 = 0 = R_1$ , the linearity of these recurrence relations guarantees that  $T_N$  and  $R_N$  are proportional to each other with a ratio of the different prefactors, that is

$$T_N = \frac{2p}{2+4p} R_N \implies C = \frac{3p}{1+2p} \quad \forall p \in [0, 1] \quad (19)$$

replicating our previous result. Fig. 4 compares the theoretical and simulated values of  $C$ , showing a high level of accuracy in our prediction. At the boundaries, the theory perfectly captures the extremes of the network topology. At  $p = 0$ , the model correctly gives  $C = 0$ , as the all-or-none model will generate a tree graph with no triangles. At  $p = 1$ , the model outputs a fully connected graph, which naturally results in  $C = 1$  since for every triangle there are three corresponding triplets. For intermediate probabilities, the clustering coefficient exhibits a non-linear, concave scaling. An initial increase from  $p = 0$  results in a rapid rise in clustering, since a small increase in connection probability significantly increases the likelihood of a new node connecting to a target node's neighbours. However, as  $p$  approaches 1 the network becomes dense and these connections become less impactful, and the increase in clustering begins to saturate.

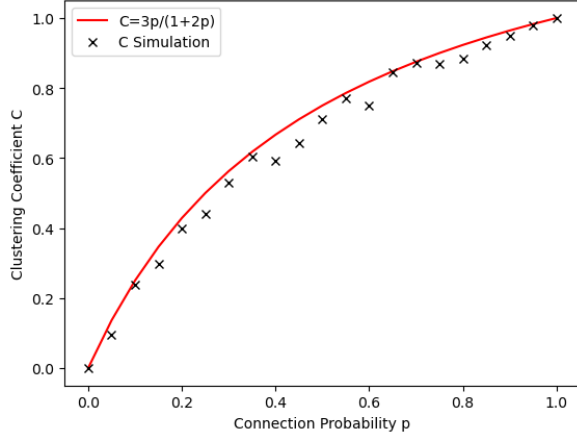


FIG. 4. Clustering coefficient plotted over range  $p \in [0, 1]$  for network size  $N = 100$ .

Assortativity is a preference for a networks' nodes to attach to others that are similar in some way. Here we examine assortativity in terms of a node's degree through the use of the Pearson correlation coefficient  $r$  of degree between a pair of linked nodes, where  $r \in [-1, 1]$ . When  $r > 0$  the network is assortative meaning nodes tend to connect to other nodes with similar degree. When  $r < 0$  the network is said to be disassortative being the opposite of assortative, and when  $r = 0$  connections are random with respect to node degree. The Pearson coefficient is given by

$$r = \frac{\sum_{j,k} jk(e_{jk} - q_j q_k)}{\sigma_q^2} \quad (20)$$

where the indices  $j$  and  $k$  denote possible values of the remaining degree of the pair of linked nodes, meaning their degrees excluding the link between them.  $e_{jk}$  is the joint probability distribution of the remaining degrees,  $q_k$  is the normalized distribution of the remaining degree and  $\sigma_q^2$  is the variance of  $q$ . To analytically estimate  $r$ , we can, as derived by Newman [1], rewrite Eq. (20) as

$$r = \frac{M^{-1} \sum_i j_i k_i - [M^{-1} \sum_i \frac{1}{2}(j_i + k_i)]^2}{M^{-1} \sum_i \frac{1}{2}(j_i^2 + k_i^2) - [M^{-1} \sum_i \frac{1}{2}(j_i + k_i)]^2} \quad (21)$$

where  $j_i$  and  $k_i$  are the degrees of the vertices at the ends of the  $i$ th edge, with  $i = 1, \dots, M$ . Noting that in previous sections we have defined  $E = M$ , and that  $\sum_i (j_i + k_i) = \sum_v k_v^2 = 2E + 2R$ , we have

$$r = \frac{2ES_1 - 2(R + E)^2}{ES_2 - 2(R + E)^2} \quad (22)$$

where we have defined  $S_1 = \sum_i j_i k_i$  and  $S_2 = \sum_i j_i^2 + k_i^2$ . We also note that  $S_2 = \sum_{v \in V} k_v^3$ , where  $V$  is the set of all vertices, from which we can derive the recurrence relation

$$S_{2,N} - S_{2,N-1} = \frac{4p}{N-1} S_{2,N-1} + \frac{6+12p}{N-1} R_{N-1} \quad (23)$$

$$+ \frac{12+20p}{N-1} E_{N-1} + 2 \quad (24)$$

which in the limit  $N \rightarrow \infty$  gives

$$S_{2,t} = \begin{cases} c_1 t + c_2 t^{2p} + c_3 t^{3p} + c_4 t^{4p} & \text{if } p \neq 0, \frac{1}{4}, \frac{1}{3}, \frac{1}{2} \\ 240t^2 - 576t^{\frac{3}{2}} + 338t + 74t \ln(t) & \text{if } p = \frac{1}{2} \\ 720t^{\frac{4}{3}} - 474t - 244t^{\frac{2}{3}} - 300t \ln(t) & \text{if } p = \frac{1}{3} \\ -1112t + 1296t^{\frac{3}{4}} - 182t^{\frac{1}{2}} + 252t \ln(t) & \text{if } p = \frac{1}{4} \\ 26t - 12 \ln(t) - 24 & \text{if } p = 0 \end{cases} \quad (25)$$

where  $c_1, c_2, c_3$  and  $c_4$  are algebraic functions exclusively dependant on  $p$ . We can also show that

$$S_{1,N} - S_{1,N-1} = \frac{4p}{N-1} S_{1,N-1} + \frac{3p}{N-1} T_{N-1} + 1 \quad (26)$$

$$+ \frac{4+12p}{N-1} E_{N-1} + \frac{2+6p}{N-1} R_{N-1} \quad (27)$$

which in the limit  $N \rightarrow \infty$  gives

$$S_{1,t} = \begin{cases} d_1 t + d_2 t^{2p} + d_3 t^{3p} + d_4 t^{4p} & \text{if } p \neq 0, \frac{1}{4}, \frac{1}{3}, \frac{1}{2} \\ 108t^2 - 258t^{1.5} + 151t + 33t \ln(t) & \text{if } p = \frac{1}{2} \\ 304t^{\frac{4}{3}} - 201t - 102t^{\frac{2}{3}} - 126t \ln(t) & \text{if } p = \frac{1}{3} \\ -449t + 522t^{\frac{3}{4}} - 73t^{\frac{1}{2}} + 102 \ln(t) & \text{if } p = \frac{1}{4} \\ 9t - 4 \ln(t) - 8 & \text{if } p = 0 \end{cases} \quad (28)$$

where  $d_1, d_2, d_3$  and  $d_4$  are again algebraic functions exclusively dependant on  $p$ . Importantly the coefficients in Eq. (25) and Eq. (28) have the relation that when  $p = 1$ ,  $c_i = 2d_i \quad \forall i = 1, 2, 3, 4$ , which in turn makes  $S_2 = 2S_1$  enforcing  $r = 1$ .

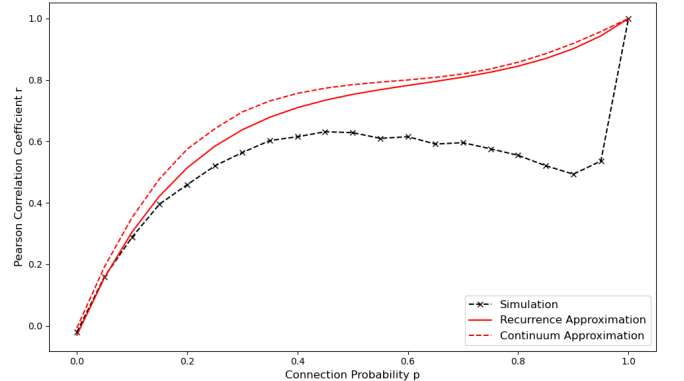


FIG. 5. Pearson correlation coefficient  $r$  against the connection probability  $p$ , for a network of size 1000.

We first note that at  $p = 1$  both approximations give precisely  $r = 1$ , matching perfectly with the simulated data. The network is perfectly assortative. This correctly describes the all-or-none model since, when  $p = 1$ , the model creates a fully connected graph, thus nodes are guaranteed to connect to all other nodes. Furthermore, when  $p = 0$ , both models correctly predict that  $r \approx 0$  again matching the simulated data. When  $p = 0$ , the network produced by the all-or-none model is a random tree graph which is correctly reflected by near-zero assortativity. From Fig. 5, we can see that for intermediate connection probabilities  $p$  both approximations capture

the general trend in how  $r$  evolves. For  $p < 0.5$  they capture the initial rapid increase in  $r$ , whilst being slight overestimates. However for  $p > 0.5$  the approximations predict a decelerating monotonic increase as the curves asymptotically approach  $r = 1$  whereas the simulated data displays a dip in  $r$  before converging with the approximations again at  $p = 1$ .

This discrepancy and overestimation arises due to the non-linearity of  $r$ . In previous sections of this report, the recurrence approximations have successfully modelled various network properties. However,  $r$  requires that we use such estimates in the non-linear Eq. (22), leading to error since the expectation of a ratio is not equal to the ratio of expectations. Since all the analytical models rely on expected values, they smooth out the natural variance of node degree, failing to account for extremely large hubs and isolated nodes generated by the randomness of the all-or-none model. By ignoring these local structures they view the network as a more uniform system, causing the overestimation in  $r$ . This sensitivity to structure is reflected in the simulated values of  $r$ , shown by the aforementioned dip. When  $p$  is large ( $p \approx 0.8$ ) the dip is most prevalent. In this regime of the model, most nodes are connected to other nodes; however, a few nodes remain very isolated. Since these isolated nodes are likely to be connected to nodes of high degree, this creates a stark contrast in degree which in turn will be penalized in  $r$ . This contrast will only be overcome when  $p$  approaches exactly one.

#### IV. AVERAGE SHORTEST DISTANCE

The average shortest-path distance between a pair of nodes is

$$L_N = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=1}^{i-1} d(v_i, v_j) \quad (29)$$

where  $d(v_i, v_j)$  is the shortest distance between node  $i$  and node  $j$ . By setting  $D_N$  equal to the double sum in Eq. (29), we can derive the following recurrence relation

$$D_N = \frac{N+1}{N-1} D_{N-1} + 1 + (N-2)(1-p) \quad (30)$$

which in the limit  $N \rightarrow \infty$  gives the continuum approximation

$$D_t = (1-p)t^2 \ln(t) + t^2 \quad (31)$$

As shown in 6, we can see the recurrence approximation perfectly aligns with the simulated data. The continuum approximation overestimates  $L$ , whilst successfully representing the overall trend of the data. This overestimation

is due to the fact we anchored the continuum approximation using the condition  $D(t=1) = 1$ . This linear decrease in  $L$  as a function of  $p$  represents the model's output accurately, going from a sparse regime to a dense regime. When  $p = 0$ , the model outputs a random tree,

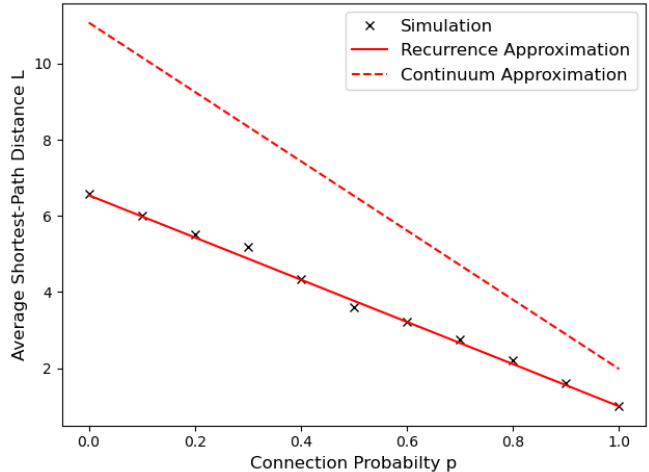


FIG. 6.  $L$  as a function of  $p$  for network size  $N = 100$ .

hence maximising  $L$ . When  $p = 1$ , the graph is fully connected hence  $L = 1$  which is perfectly predicted by our recurrence approximation. As  $p$  increases new nodes begin to attach to the neighbours of the target node, creating topological shortcuts in the network, hence decreasing  $L$ .

#### V. CONCLUSION

Through methods of approximation and numerical simulations this report found the underlying structure of the all-or-none model by varying parameters  $p$  and  $N$ . The recurrence approximations were highly effective in the estimation of the number of edges, the clustering coefficient, and the average shortest distance. Whilst approximations captured the general trend in the Pearson correlation coefficient, they overestimated the values since they smooth out natural variation and fail to account for isolated nodes created due to the random nature of the model. These findings highlight the usefulness of the all-or-none model, showing how a simple parameter such as the connection probability  $p$  can generate complex rules that accurately model the network's structure.

#### CODE AVAILABILITY

Full code used throughout report available at my GitHub repository

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- [1] M. E. J. Newman, *Assortative mixing in networks*, Phys. Rev. Lett. **89**, 208701 (2002).